

Module 1: Microsoft Excel for Data Analytics

1. Introduction to Excel

- Overview of Excel interface
- Basic operations (data entry, formatting, and basic formulas)

2. Data Manipulation

- Sorting and filtering data
- Using PivotTables and Pivot Charts
- Data validation

3. Advanced Formulas and Functions

- Logical functions (IF, AND, OR)
- Lookup functions (VLOOKUP, HLOOKUP, INDEX, MATCH)
- Text functions and date functions

4. Data Analysis Tools

- Using Data Analysis Tool Pak
- What-If Analysis (Scenario Manager, Goal Seek, Data Tables)

5. Visualization

- Creating charts and graphs
- Conditional formatting
- Sparklines and data bars

Module 2: Power BI

1. Introduction to Power BI

- Overview of Power BI tools (Desktop, Service, Mobile)
- Understanding data sources and data types

2. Data Import and Transformation

- Connecting to various data sources
- Data cleaning and transformation using Power Query Editor

3. Data Modeling

- Creating relationships between tables
- Understanding DAX (Data Analysis Expressions)
- Calculated columns and measures

4. Visualization

- Creating and customizing visuals (charts, maps, tables)
- Using slicers and filters
- Designing interactive dashboards

5. Sharing and Collaboration

- Publishing reports to Power BI Service

- Creating and sharing dashboards
- Setting up data refresh and alerts

Module 3: Transact-SQL (T-SQL)

1. Introduction to T-SQL

- Understanding SQL Server and T-SQL
- Basic SQL syntax and commands

2. Data Querying

- SELECT statements
- Filtering data with WHERE clause
- Sorting and limiting results

3. Data Manipulation

- INSERT, UPDATE, DELETE statements
- Using joins to combine data from multiple tables

4. Advanced Querying

- Subqueries and nested queries
- Common Table Expressions (CTEs)
- Window functions

5. Stored Procedures and Functions

- Creating and executing stored procedures
- User-defined functions
- Error handling and transactions

Module 4: Python Basics

1. Introduction to Python

- Setting up the environment (Anaconda, Jupyter Notebook)
- Basic syntax and data types
- Control flow statements (if, for, while)

2. Data Structures

- Lists, tuples, sets, dictionaries
- List comprehensions and generator expressions

3. Functions and Modules

- Defining and calling functions
- Importing and using modules
- Error and exception handling

Durations 85 hrs.

Module 5: Python for Data Analysis

1. NumPy

- Introduction to NumPy arrays
- Array operations and broadcasting
- Working with multidimensional arrays

2. Pandas

- Introduction to pandas Data Frames and Series
- Data manipulation with pandas (filtering, merging, grouping)
- Handling missing data

3. Matplotlib

- Introduction to data visualization with Matplotlib
- Creating basic plots (line, bar, scatter)
- Customizing plots (titles, labels, legends)

4. Seaborn

- Introduction to Seaborn
- Creating statistical plots (boxplot, violin plot, pairplot)
- Customizing Seaborn plots

Module 6: Capstone Project

1. Project Planning

- Defining the problem statement
- Identifying data sources and requirements

2. Data Collection and Preparation

- Collecting and cleaning data
- Exploratory data analysis

3. Data Analysis and Visualization

- Analyzing data using Excel, Power BI, T-SQL, and Python
- Creating visualizations to present findings

4. Report and Presentation

- Preparing a final report
- Presenting the project findings